

GCP Hands-On:

Hands-on experience of core Google Cloud features.

Observability:

Module 4: Proactive Observability & Health Checks

File Name: `04-observability/README.md`

Objective: Implement automated, global uptime monitoring to ensure the Cloud Function is accessible and responding correctly to HTTP requests.

1. Monitoring Configuration (Infrastructure as Code)

To ensure a reproducible setup, the Uptime Check was defined using a JSON payload and deployed via the Google Monitoring API. This monitors the function from multiple global locations every 60 seconds.

Config Definition:

JSON

```
{
  "displayName": "Hello Function Health",
  "monitoredResource": {
    "type": "uptime_url",
    "labels": {
      "host": "asia-southeast1-[PROJECT_ID].cloudfunctions.net"
    }
  },
  "httpCheck": {
    "requestMethod": "GET",
    "path": "/hello-pubsub-func",
    "port": 443,
    "useSsl": true
  },
  "period": "60s"
}
```

2. Technical Pivot: Bypassing CLI Abstractions

During deployment, several issues were encountered with the `gcloud monitoring` CLI's flag parsing and case sensitivity.

- **The Problem:** Inconsistent flag support for `--display-name` and `--period` across SDK versions.

- **The Solution:** Switched to a direct **REST API call** via **cURL**. This bypassed the CLI wrapper to deliver a precise payload to the Monitoring V3 API.

Bash

Direct API Deployment

curl -X POST \

"https://monitoring.googleapis.com/v3/projects/\${PROJECT_ID}/uptimeCheckConfigs" \

-H "Authorization: Bearer \$(gcloud auth print-access-token)" \

-H "Content-Type: application/json" \

-d @uptime-config.json

3. Verification & Metrics

The deployment was verified by checking the **Uptime Checks Dashboard**.

- **Initial Status:** **Pending** (Grey icon) while global checkers propagate.
 - **Final Status:** **Passed** (Green icon) indicating the function is reachable from 100% of global testing locations.
-

Root Level: Master Project Documentation

File Name: **README.md** (at the root of your repo)

Google Cloud Platform: Core Foundations Lab

A hands-on repository documenting the deployment, security, and observability of an event-driven cloud ecosystem on GCP.



Overview

This project serves as a technical log of core GCP services integrated to form a functional infrastructure. It emphasizes a **CLI-first** approach and **SRE (Site Reliability Engineering)** principles for troubleshooting.



Project Modules

1. **[01-Cloud Storage](#)**: Object storage, bucket policies, and public access logic.
2. **[02-Event-Driven Pub/Sub](#)**: Asynchronous messaging and service decoupling.
3. **[03-Serverless Logic](#)**: Deploying 2nd Gen Cloud Functions with specific IAM role resolutions.

4. [04-Observability](#): Automated health checks and direct REST API interaction.

Key Technical Skills

- **IAM Troubleshooting:** Resolved Cloud Build bottlenecks by assigning `serviceAccountUser` and `artifactregistry.writer` roles.
- **REST API Interaction:** Utilized `cURL` and JSON payloads to bypass CLI version discrepancies.
- **Cloud Architecture:** Implemented an event-driven flow: `Pub/Sub` -> `Cloud Function` -> `Logic Execution`.

Set up your project's observability

0% complete

Alerts

0

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Services with SLO out of budget

0

[View services](#)

Failed uptime checks

0

[View checks](#)

VMs with ops agent

0 of 0

No legacy agents. [View VMs](#)

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Alerts

0

[View alerts](#)

Services with SLO out of budget

0

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Failed uptime checks

0

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VMs with ops agent

0 of 0

No legacy agents. [View VMs](#)Predefined ▼< 🕒 Last 1 hour ▼ IST 🔍 > 🗖 ✎ Customise

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Google Cloud Observability offers a full suite of features for monitoring and troubleshooting your workloads, from logs, dashboards, alerts and traces to SLO and uptime checks. You can customise this page to quickly access the views that matter most to you.

Logs

An at-a-glance view of the health of your applications. [View all logs here.](#)

Widget

Compute Engine logs by severity ▼

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Google Cloud

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Monitoring

☆

Uptime checks

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Logs Explorer

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Error reporting

Uptime checks

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Release notes

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Display Name ↑

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Europe

North America

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Policies

No rows to display



Uptime checks



+ Create uptime check

+ 1 recommended alert



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Display Name ↑	Asia Pacific	Europe	North America	South America	
Hello Function Health	●	●	●	●	⋮